COSC 461/561 Project 1: Eagle Delivery Report

Part A:

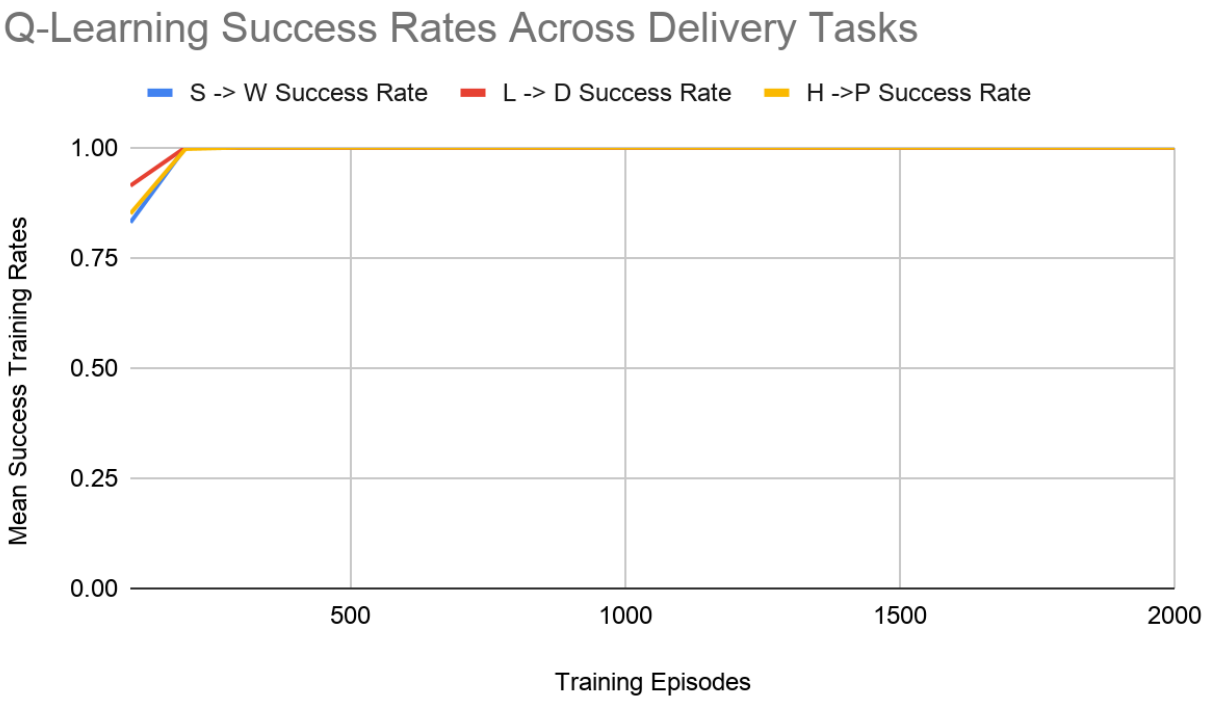
- **Formulate the problem**
  - The initial state of the program is the starting building. If the robot begins in Halle Library, then the initial state is L; if it begins in Downing Hall, then it is D; and so on
  - The state of the program is the robot's current location. Any traversable path or building can represent the current state.
  - The robot has four possible actions: Up, Down, Left, and Right.
  - The environment is a deterministic environment, meaning that the search algorithm is fully aware of it. This means that every action and its result is already known in advance
  - The goal test checks whenever the robot has reached the entrance of a building set as the goal.
  - The cost of each path is always 1.
  - Because the cost of each path is 1 regardless of failed or successful movements, BFS becomes the most optimal search algorithm because it searches each state level by level. This means that theoretically, BFS will have found the shortest cost path the first time it reaches a goal.
- **TODO 1: Manhattan**
  - The Manhattan distance is admissible because it can't overestimate the true shortest path cost.
- **BFS and A* Comparison**

Map	Delivery	BFS Length	A* Length	BFS Expanded	A* Expanded
Base	S -> W	18	18	67	22
Base	L -> D	22	22	79	62
Base	H -> P	16	16	41	16
Detour	S -> W	26	26	70	51
Detour	L -> D	22	22	65	60
Detour	H -> P	16	16	41	16

  - The length of the routes in both BFS and A* are the same across each start and goal, but the expanded nodes are different. A* expanded less nodes on every occasion. However, the difference between expanded nodes between BFS and A* is not consistent; the advantage fluctuates between a big advantage and a minor advantage.

Part B:

- **Results and Interpretations**
  - When checking each generated CSV file, I've noticed that the mean success rates increased very quickly after the first tests. All delivery tasks eventually reached 100% success rate by at most the 300th training episode. Afterward, the success rate stabilized and stayed at 100%



Part C:

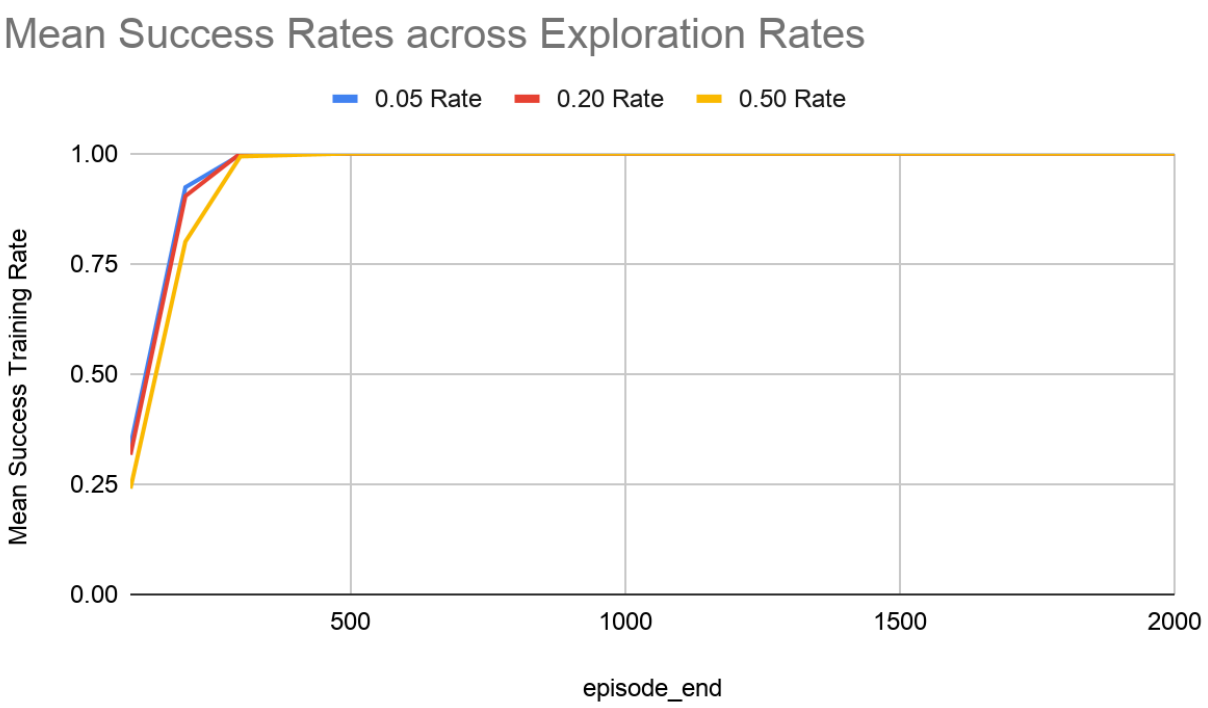
- **Personal Experimentation and Question**
  - Question: My question is to explore how the exploration can affect the learning and performance
  - My prediction before testing: A lower exploration rate may result in a slower learning rate because the agent will likely not encounter or discover better routes earlier in the training.
- **Custom Map**
  - Loop: There is a loop near the center of the map, which opens up multiple routes to several destinations
  - Dead End: At the lower left corner of the map, there is a dead end. If the agent ever reaches this, it must backtrack.
- **BFS and A* Search on Custom Map**

Search	Expanded	Path Length
BFS	81	34

A*	74	34

- **Experiment Setup**
  - Only the exploration rate was changed. The learning rate, discount, and slip probability remained the same. The training episodes, evaluation episodes, and random seeds are also the same

Exploration	Success Rate	Mean Return	Mean Steps
0.05	1	6	34
0.20	1	6	34
0.50	1	6	34



- **Results Discussion**
  - While the Success Rate, Mean Return, and Mean Steps eventually became the same across all exploration rates, we can see in the line chart that the mean success rate varies between the exploration rates. Notably, it took 0.20 rate only 300 episodes to reach 100% success rate whereas it took 0.05 and 0.50 rate 400 and 500 episodes respectively to reach 100%.
  - The results I received both prove yet contradict my hypothesis. While 0.20 did prove to have a faster learning rate than 0.05, the results for 0.50 showed it had a slower learning rate than 0.05 and 0.50.
    - This leads me to believe that while an increase of exploration rate can improve learning rate, there will be a point where a higher exploration rate will eventually lead to a slower learning rate.